Kavin Balaji S

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Education

Bachelor of Technology in Computer and Communication Engineering *2023-2027*
Amrita Vishwa Vidyapeetham University, Coimbatore CGPA: 7.11

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Machine Learning, DBMS, Embedded Systems, IoT and Cloud Computing, Computer Networks

Experience

IoT Engineer *July 2024 - October 2025*
Intel IoT Club, Amrita Vishwa Vidyapeetham University

- Conducted hands-on workshops and technical sessions, introducing members to IoT concepts and brainstorming ideas.

Technical Skills

Programming Languages	C, Python, JavaScript, HTML
Frameworks & Libraries	React.js, Node.js, Express.js, TensorFlow, Scikit-Learn, Librosa, Pandas, NumPy
Tools & IDEs	VS Code, Keil uVision, Arduino IDE, LogiSim, Postman, GNU Radio, NetSim
Cloud & Databases	AWS, MySQL, MongoDB
Embedded & IoT	Arduino, ESP32, MQTT, Node-RED, Blynk
Version Control	Git, GitHub
Interests	Web Development, IoT Systems, Cloud Computing

Projects

StartSmart – AI Startup Evaluator | *MongoDB, Express.js, React.js, Node.js, GPT 4o API* GitHub • Demo

- Built a startup funding platform connecting entrepreneurs with investors, featuring AI-powered idea analysis, automated funding requests, real-time negotiations, and comprehensive notification systems.
- Designed role-based dashboards with portfolio analytics, market research tools, ideathon competition management, idea comparisons, and admin analytics for monitoring platform growth and user engagement.

Smart Appliance Manager | *C++, MQTT, Node-RED, Blynk* GitHub

- An IoT-based home automation system to automatically control and monitor appliances in real time.
- Built with ESP32 microcontroller for automatic temperature and ambient light based appliance control and monitoring.
- Implemented MQTT protocol communication with Node-RED and Blynk dashboards for remote monitoring and real-time data visualization.

Music Genre Classification | *Python, Librosa, scikit-learn, TensorFlow* GitHub

- Built a music genre classifier extracting features (MFCCs, chroma, spectral contrast) using Librosa.
- Optimized accuracy by training SVM, Random Forest, and CNN models.
- Achieved high prediction performance validated by confusion matrices and evaluation metrics.

Library Management System | *Python, Tkinter, Pandas, OpenPyXL* GitHub

- An application made using Python for managing all facets of library operations.
- Designed and implemented book cataloging, user registration, borrowing, returns with overdue fee calculation, and reservation workflows.
- Persisted data to Excel files and logged all transactions.
- Developed an intuitive multi-frame GUI for both administrator and user modes.

Certifications

AWS Certified Cloud Practitioner CLF-C02 *Verify*